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Explainable AI for Multi-Label Chest X-ray Diagnosis: Layer-wise Grad-CAM with Hierarchical Feature Extraction

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***Abstract*— Artificial intelligence (AI) has become indispensable in medical image analysis, with models such as convolutional neural networks (CNNs) and Transformer achieving remarkable success in diagnostic imaging. Despite their impressive performance, these models often lack interpretability, limiting their adoption in clinical workflows where understanding disease-specific features is critical for trust.**

In this study, we propose an explainability framework that enhances interpretability for multi-label disease classification in chest X-ray (CXR) diagnosis by utilizing the U-Net encoder-decoder architecture. The encoder and decoder outputs are concatenated to effectively capture hierarchical features for the classification of 14 observations in the MIMIC-CXR dataset. To further improve interpretability, we apply gradient-weighted class activation mapping (Grad-CAM) across multiple layers, providing detailed insights into the refinement of hierarchical features and the emphasis on disease-specific regions throughout the network. This integration of U-Net with an explainable AI (XAI) framework enhances transparency in the diagnostic process, supporting more informed and trustworthy clinical decision making.

***Clinical relevance*— This study underscores the importance of interpretability in AI-based radiology. By providing clear Grad-CAM visualizations of disease-specific features, clinicians can more confidently validate model predictions and incorporate these insights into their decision-making processes. Through enhanced transparency, our approach not only improves diagnostic performance, but also fosters greater trust in AI tools, paving the way for these models to serve as robust, clinician-friendly decision support systems in routine radiological workflows.**

I. INTRODUCTION

Rapid advancement of machine learning has significantly transformed medical imaging, particularly through the application of AI models such as convolutional neural networks (CNNs) and Transformer-based architectures [1], [2]. These models have achieved remarkable performance in classifying multiple disease observations in chest X-rays (CXR), demonstrating their potential to support clinicians in diagnostic workflows and reduce workload.

However, high diagnostic accuracy alone is not sufficient in clinical practice [3]. Clinicians require not only reliable predictions but also interpretable insights that align with medical reasoning. Deep learning models are often criticized for their opacity, functioning as black boxes with limited transparency in decision-making [4], which hinders their integration into real-world medical workflows.

Explainable AI (XAI) techniques aim to address this limitation by providing visual and feature-level explanations that highlight important input regions [4]. Among them, gradient-weighted class activation mapping (Grad-CAM) [5] is widely adopted in medical imaging due to its ability to produce visual explanations without modifying the model architecture. Prior studies have used Grad-CAM to identify disease-relevant regions in CXR diagnosis models [1], [6].

Despite its utility, most Grad-CAM applications focus only on final-layer visualizations, overlooking how features evolve across the network. In medical image classification, hierarchical feature extraction is crucial, with early layers capturing low-level textures and deeper layers encoding high-level semantics. Without layer-wise interpretability, it remains unclear how and when disease-specific features emerge during inference. This highlights the need for a more comprehensive framework that reveals feature refinement throughout the encoder-decoder architecture.

To address this, we propose a multi-label classification model for 14 observations using the MIMIC-CXR dataset [7], [8], [9], labeled via the CheXpert NLP tool [10]. We adopt U-Net [11], a well-established encoder-decoder architecture known for its multiscale feature extraction. Grad-CAM is applied at each layer of the U-Net model to visualize how features are refined and how disease-specific regions are emphasized throughout the network. Unlike conventional Grad-CAM methods that visualize only the final layer, our approach captures the progressive emergence of disease-relevant features across encoder and decoder layers. This framework bridges the gap between interpretability and diagnostic reliability, promoting more transparent and trustworthy AI in medical imaging.

II. RELATED WORK

The use of AI models in medical diagnosis is gaining interest, particularly in disease classification tasks using chest radiograph (CXR) images. These models can be categorized

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into CNN-based and Transformer-based approaches. CNN-based models, such as CheXNet [12], have achieved high precision in detecting observations such as pneumonia by capturing localized features in medical images. Transformer-based models, including the Vision Transformer (ViT) [13] and the Swin Transformer [14], effectively capture both local and global features, making them suitable for complex diagnostic tasks [2]. However, both architectures face a common limitation: the lack of explainability, which complicates clinical adoption by making it difficult to interpret the rationale behind predictions. This has led to increased research on explainable AI (XAI) to enhance transparency and trust in AI-assisted diagnosis.

XAI techniques address this by revealing decision-making processes in deep models. These techniques, including LIME [15], SHAP [16], Grad-CAM [5], and LRP [17], help interpret model predictions by identifying key features or regions. Among them, Grad-CAM is particularly popular in image analysis, generating visual explanations without requiring modifications to the model architecture. In the medical field, particularly for the classification of CXR disease, Grad-CAM and other XAI methods have been applied to improve clinical trust by highlighting relevant regions in diagnostic images [4], [1]. However, many of these methods are limited to simple visualizations and do not provide detailed analyses of how features evolve across network layers. To overcome this limitation, further research is needed to apply Grad-CAM in a layer-wise manner, enabling deeper insights into feature representation and improving interpretability.

III. METHODS

A. Introduction of the proposed model

The proposed model modifies the U-Net architecture [11] to support multi-label disease classification, as illustrated in Fig. 1. Although U-Net is traditionally designed for segmentation tasks, it has been adapted here for classification by incorporating several adjustments. Specifically, the encoder is replaced with pre-trained ResNet blocks [18], which facilitate hierarchical feature extraction through residual connections. This improves the model's capacity to capture complex patterns present in chest X-ray (CXR) images. The choice of ResNet was motivated by its robust performance in medical imaging applications, its stability in deep feature extraction through residual connections, and its relatively simple architecture that facilitates clear layer-wise interpretability compared to more complex models such as DenseNet or Transformer-based backbones. Skip connections transfer features from the encoder to the corresponding decoder layers, where up-sampling operations reconstruct detailed feature representations.

To further utilize both encoder and decoder information, the model concatenates their final outputs to create a unified feature representation. This combined output is processed through global average pooling (GAP) and passed to a fully connected layer for the prediction of multiple disease labels. Additionally, Grad-CAM visualizations are applied across the network to analyze how hierarchical features contribute to the classification task, offering interpretability to support clinical decision-making.

B. Dataset and Preprocessing

The dataset used in this study is MIMIC-CXR [7], [8], [9], a publicly available collection of 377,110 chest radiographs paired with clinical reports. To derive consistent diagnostic labels, we employed the CheXpert NLP tool [10], which extracts and categorizes radiological findings into 14 observations. These observations, along with brief clinical descriptions, are summarized in Table I. All labels were simplified into binary classes (positive = 1, negative = 0).

TABLE I
CHEXPERT-LABELED 14 OBSERVATIONS AND THEIR CLINICAL DESCRIPTIONS

Label	Description
Atelectasis	Partial or complete collapse of lung tissue
Cardiomegaly	Enlargement of the heart silhouette
Consolidation	Alveolar filling (e.g., due to pneumonia)
Edema	Excess fluid in lung tissues
Enlarged CM	Enlarged Cardiomediastinum
Fracture	Bone break or crack in rib or chest structure
Lung Lesion	Focal lung abnormality (e.g., tumor, granuloma)
Lung Opacity	General opacities in the lung fields
No Finding	Normal appearance; no evident pathology
Pleural Effusion	Fluid accumulation in the pleural space
Pleural Other	Other pleural abnormalities
Pneumonia	Infection causing lung inflammation
Pneumothorax	Presence of air in the pleural space
Support Device	Presence of devices (e.g., pacemaker, lines)

To maintain consistency in the imaging perspective, only Posteroanterior (PA) and Anteroposterior (AP) view images were included, while lateral views were excluded. All images were resized to 224×224 pixels for computational efficiency and standardized input dimensions. A metadata DataFrame linking each StudyID to its diagnostic labels and image filenames was constructed for efficient data handling during training and evaluation.

To ensure computational feasibility and focus the analysis, 100,000 samples were randomly selected from the preprocessed dataset. This subset retained the original label distribution of the entire dataset, ensuring a balanced representation of diagnostic categories while reducing computational overhead.

The MIMIC-CXR dataset used in this study was collected in accordance with institutional review board (IRB) approval and has been made publicly available for research use. No new human subject data were collected for this study.

C. Model Training & evaluation setup

After preprocessing and sampling, the dataset was split into training, validation, and test subsets with a ratio of 70:15:15. The training set was used for model optimization and the validation set guided hyperparameter tuning and

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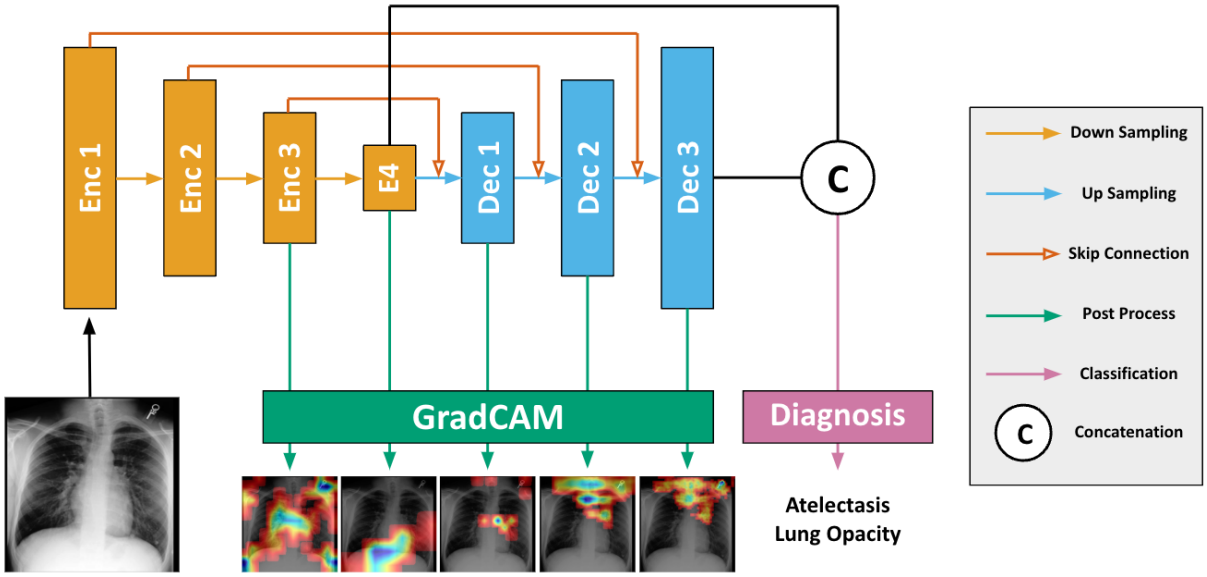


Fig. 1. **Workflow diagram of the proposed U-Net-based diagnostic model with integrated Grad-CAM for explainability** The encoder, built with pre-trained ResNet blocks, extracts hierarchical features from the input chest X-ray (CXR) image. Skip connections pass features from the encoder to the corresponding decoder layers, where the decoder reconstructs detailed representations using up-sampling operations. Unlike conventional U-Net architectures used for segmentation, this model adapts the U-Net structure for multi-label classification by concatenating the final encoder and decoder outputs. This concatenated feature representation is then processed through global average pooling (GAP) and a fully connected classification head, enabling the model to make multi-label predictions. Grad-CAM is applied to visualize the contribution of each layer to the decision-making process, highlighting disease-specific regions in the CXR. The arrows represent down-sampling (orange), up-sampling (blue), and Grad-CAM processing (green).

early stopping. The test set was evaluated using AUC, a metric chosen for its clinical relevance and its ability to assess model performance independently of specific thresholds.

To evaluate the diagnostic performance of the proposed U-Net model against a baseline model, we trained a baseline architecture using only a pre-trained CNN backbone (detailed in Table II). This comparison enabled a study of how the encoder-decoder structure in U-Net influences classification performance.

D. Grad-CAM Application

Grad-CAM (Gradient-weighted Class Activation Mapping) is an explainable AI (XAI) technique that helps users visualize which regions of an input image most strongly influence predictions of a deep learning model [5]. In essence, Grad-CAM highlights the areas in an image that the model considers important when determining a specific class or label. It does so by computing the gradient of the target class score with respect to the feature maps of a chosen convolutional layer, then weighting and summing those gradients to create a heatmap. This heatmap can be overlaid on the original image, revealing the spatial regions that contributed the most to the final prediction.

In this study, Grad-CAM was applied to our U-Net-based architecture (Fig. 1) to improve interpretability in a multi-label diagnostic setting. Unlike single-label tasks, where only one observation is predicted at a time, multi-label classification requires the model to identify multiple potential observations within a single image. We therefore applied Grad-CAM to both the encoder and decoder layers of the U-Net to understand how different parts of the image inform each disease label. By backpropagating gradients from the classification head to each layer's activation maps, we generated heatmaps that reveal disease-relevant features at various stages of the network. This layer-wise approach offers granular insights, enhancing interpretability and clinical trust.

IV. RESULTS

A. Diagnostic Performance of the U-Net Model

The diagnostic performance of the Base CNN and U-Net models across 14 diagnostic categories is summarized in Table III. The Area Under the Curve (AUC) scores provide a detailed comparison of the ability of the two architectures to distinguish between positive and negative cases for each observation.

In general, both models demonstrated similar diagnostic capabilities. For example, "Pleural Effusion" achieved an AUC score of 0.897 with the U-Net model and 0.893 with the Base CNN model, while "Edema" showed scores of 0.880 and 0.879, respectively. These results suggest that both architectures are effective in identifying observations where precise localization and feature extraction are critical. In some cases, the Base CNN model performed slightly better, such as for "Fracture" (AUC = 0.682 vs. 0.663) and "Pneumothorax" (AUC = 0.855 vs. 0.851). This indicates that simpler architectures can still achieve competitive performance under certain observations.

In contrast, the AUC scores for observations such as "Fracture" (0.663) and "Lung Lesion" (0.714) were comparatively lower for both models. Labels such as "Pleural

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TABLE II
MODEL DESCRIPTION AND COMPARISON

Model	U-Net	Base CNN
Architecture Type	U-Net (Encoder-Decoder)	ResNet34 (Encoder Only)
Encoder Backbone	Pre-trained ResNet34	Pre-trained ResNet34
Decoder Structure	3 Decoder Layers ConvTranspose2d + Conv Blocks	None
Classification Head	Global Avg Pool (Encoder + Decoder) Dropout + Fully Connected	Global Avg Pool (Encoder Only) Dropout + Fully Connected
Dropout Rate	0.3	0.3
Number of Parameters	~23M	~21.3M
Feature Attribution	Grad-CAM on Encoder + Decoder Layers	Grad-CAM on Encoder Layers Only

TABLE III
AUC SCORES FOR BASE CNN AND U-NET MODEL

Label	Base CNN	U-Net
Atelectasis	0.798	0.801
Cardiomegaly	0.797	0.803
Consolidation	0.802	0.806
Edema	0.879	0.880
Enlarged CM	0.741	0.732
Fracture	0.682	0.663
Lung Lesion	0.717	0.714
Lung Opacity	0.741	0.742
No Finding	0.847	0.853
Pleural Effusion	0.893	0.897
Pleural Other	0.849	0.831
Pneumonia	0.711	0.715
Pneumothorax	0.855	0.851
Support Device	0.887	0.889

AUC scores greater than 0.85 are underlined, indicating observations selected for interpretability analysis.

Effusion," "Edema," and "Pneumothorax," which achieved AUC scores above 0.85, were selected for further analysis to evaluate the interpretability of the decision-making process of the U-Net model. In Table III, these labels are highlighted with an underscore to indicate their selection for Grad-CAM evaluation.

B. Grad-CAM Visualizations Analysis

Based on the results of the diagnostic performance, more analysis was performed to explore the interpretability of the model predictions using Grad-CAM. Grad-CAM visualizations were performed exclusively for disease labels with AUC scores greater than 0.85. This approach aimed to evaluate the model's decision-making process for observations in which its diagnostic performance was reliable, therefore aligning interpretability with clinical trust. In Table III, labels with AUC scores greater than 0.85 are highlighted with underlining, indicating the observations selected for further interpretability analysis.

The label "No Finding", representing normal cases, was excluded from the Grad-CAM analysis, as it does not correspond to a specific disease observation. By focusing on disease-specific labels, this study ensures that the interpretability analysis remains clinically meaningful and directly applicable to diagnostic tasks.

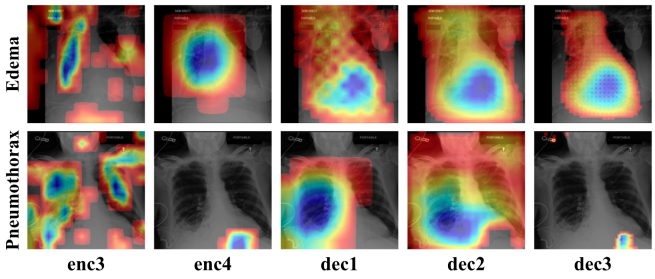


Fig. 2. **Layer-wise Grad-CAM Analysis** Grad-CAM visualizations for "Edema" and "Pneumothorax" across encoder layers 3 and 4 and decoder layers 1, 2, and 3. The heatmaps illustrate the layer-wise evolution of feature attribution, highlighting the hierarchical processing of disease-specific features in the U-Net architecture.

*1) **Layer-wise Grad-CAM Analysis**:* This analysis aims to reveal how the U-Net model refines and consolidates disease-specific features across different layers, providing insights into the hierarchical feature extraction process. Grad-CAM visualizations were extracted for both the encoder and decoder layers of the U-Net model to better understand the model decision-making process. This layer-wise analysis provided valuable insight into how the model processes disease-specific features at varying levels of abstraction.

In Fig. 2, representative Grad-CAM visualizations for "Edema" (AUC = 0.880) and "Pneumothorax" (AUC = 0.851) illustrate the differences in feature attribution across encoder layers 3 and 4 and decoder layers 1, 2, and 3. These results demonstrate how the focus of the model evolves between layers, highlighting the hierarchical nature of feature extraction in the U-Net architecture.

For "Edema", the Grad-CAM heatmaps revealed relatively consistent patterns across deeper decoder layers, suggesting that the model effectively consolidates feature information at these stages. In contrast, for "Pneumothorax", a striking inversion in Grad-CAM activation regions was observed at decoder layer 2, indicating that the model adjusts its focus dynamically depending on the disease-specific characteristics. These findings emphasize the interpretability benefits

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of the U-Net encoder-decoder structure, helps understanding of how features contribute to diagnostic predictions.

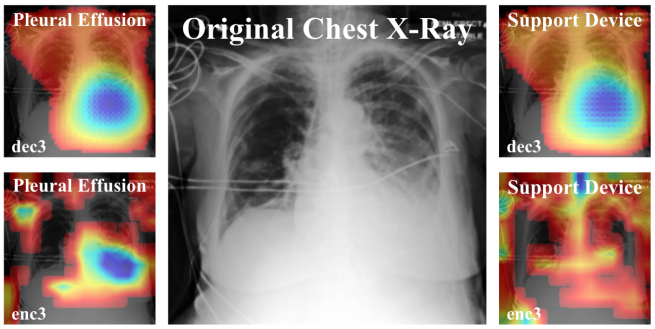


Fig. 3. **Label-wise Grad-CAM Analysis** Grad-CAM results for a patient diagnosed with both "Pleural Effusion" (left) and "Support Device" (right). Distinct activation patterns in encoder layer 3 demonstrate the model's ability to capture disease-specific features, while decoder layers show overlapping activations.

*2) **Label-wise Grad-CAM Analysis**:* This analysis explores feature activation patterns for multiple disease labels within the same layer, offering insights into how the model highlights label-specific regions. In Fig. 3, Grad-CAM results are presented for a patient diagnosed with both Pleural Effusion and Support Device. Encoder layer 3 (enc3) exhibited distinct activation patterns for the two labels: "Pleural Effusion" primarily highlighted regions corresponding to fluid accumulation in the lung base, while "Support Device" focused on localized areas associated with the physical presence of the device. In contrast, decoder layer 3 (dec3) demonstrated similar activation patterns for both labels, suggesting that the decoder layers integrate features into a more generalized representation.

This contrast between encoder and decoder layers suggests potential differences in their roles: the encoder appears to capture more disease-specific features, while the decoder may consolidate these features for a unified prediction. However, further investigation is needed to determine whether this pattern is generalized in different observations. These observations suggest that analyzing layer-wise variations in Grad-CAM outputs may provide insights into how different layers contribute to the diagnostic process and facilitate targeted model refinement for multi-label classification tasks.

*3) **Confusion Matrix-based Analysis**:* The purpose of this analysis is to evaluate the consistency and reliability of model predictions by comparing Grad-CAM visualizations across different prediction outcomes. Fig. 4 compares Grad-CAM visualizations for True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) cases for the "Pleural Effusion" label. Notably, TP cases showed clear and localized activation in disease-relevant regions, while FP cases exhibited diffuse or misplaced activation, indicating potential overfitting to non-disease-related features. TN cases displayed minimal activation, as expected, while FN cases revealed either weak or incorrectly localized activations, highlighting areas where the model struggled to identify subtle disease features.

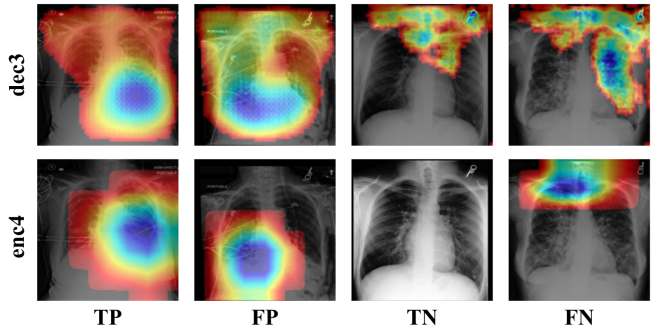


Fig. 4. **Confusion Matrix-based Analysis** For the Pleural Effusion label, Grad-CAM visualizations for TP, FP, TN, and FN cases. The results highlight differences in activation patterns, with TP cases showing localized activations and FP/FN cases revealing potential areas for refinement.

V. DISCUSSION

This study highlights the effectiveness of integrating a U-Net-based diagnostic framework with Grad-CAM for multi-label chest X-ray classification. By leveraging Grad-CAM at multiple layers, our approach enhances interpretability while maintaining diagnostic reliability, providing insights into the hierarchical nature of feature extraction in deep learning models.

A. Comparing U-Net and CNN: Performance and Interpretability

The comparison between U-Net and a conventional CNN revealed that both architectures achieved comparable diagnostic performance (Table III). While U-Net did not show a significant advantage in AUC scores, its encoder-decoder structure allowed for enhanced interpretability through Grad-CAM, making it a valuable tool for visualizing disease-specific regions. Our findings suggest that the encoder primarily captures disease-relevant features, while the decoder consolidates them into a more generalized representation. However, the role of the decoder in classification accuracy remains unclear and warrants further investigation.

B. Challenges in Multi-Label Classification

Variability in model performance across different disease labels, particularly for observations such as "Fracture" and "Lung Lesion," highlights the inherent challenges of multi-label classification. These challenges stem from class imbalance, overlapping pathological features, and subtle morphological differences that make certain observations harder to distinguish. Addressing these issues through label-specific training strategies and data augmentation could improve overall robustness.

C. Insights from Grad-CAM Analysis

The layer-wise Grad-CAM analysis provided a deeper understanding of how hierarchical features contribute to classification. While deeper decoder layers exhibited stable feature consolidation for observations like "Edema," diseases such as "Pneumothorax" showed shifting activation patterns, suggesting dynamic feature adaptation. Additionally, false

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positive and false negative cases in the confusion matrix analysis revealed potential areas for refinement, such as improving sensitivity to subtle abnormalities and reducing overfitting to non-disease regions.

D. Future Directions

Future directions will focus on optimizing the U-Net architecture to better integrate segmentation capabilities for improved spatial localization and classification, as well as generalizing the layer-wise analysis. Modifications to both the encoder and decoder structures, along with refinements in training strategies, will be explored to fully leverage U-Net's ability to highlight disease-specific regions, thereby enhancing both diagnostic accuracy and interpretability.

In addition, the layer-wise Grad-CAM analysis introduced in this study will be expanded to larger and more diverse datasets, and its adaptability across various disease observations and model architectures will be evaluated. These efforts are expected to improve the robustness and clinical utility of the proposed framework.

Moreover, while the current study focused on validating feasibility within a ResNet-based U-Net, future work will extend the framework to a wider range of state-of-the-art CXR diagnostic models, including Transformer-based architectures. Evaluating generalizability across different backbones will further clarify the strengths and limitations of layer-wise interpretability. Finally, applying the framework to longitudinal chest X-ray series could enable monitoring of disease progression or treatment response over time, offering new opportunities for dynamic clinical decision support.

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